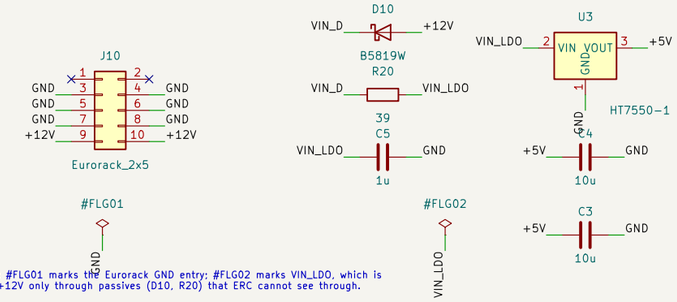
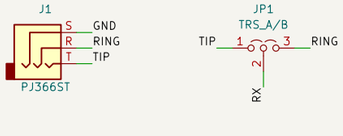
BIG MODULAR MIDI — S4 schematic (5 V logic, BM V1 direct input). Circuit intent: docs/research.md

POWER: Eurorack +12V -> B5819W -> 39R -> HT7550-1 -> +5V. -12V (J10 pins 1-2) unused.



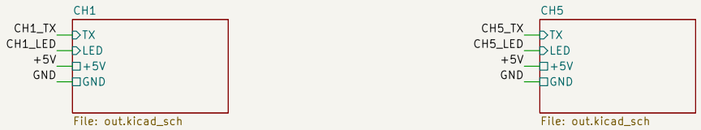
PWR_FLAG: #FLG01 marks the Eurorack GND entry; #FLG02 marks VIN_LDO, which is fed from +12V only through passives (D10, R20) that ERC cannot see through.

MIDI IN: Bread Modular V1 direct input (3.3 V and 5 V senders), no opto

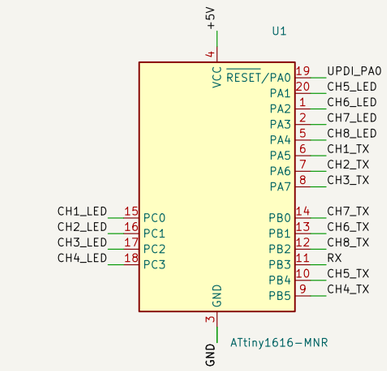


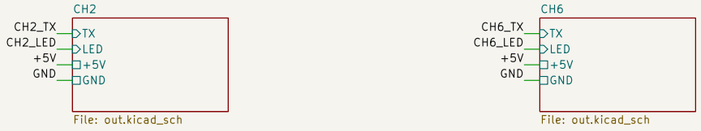
JP1 TRS TYPE (2.54 mm header on the back): shunt 1-2 = TYPE A (Tip = data), shunt 2-3 = TYPE B (Ring = data). The unused conductor is left open. Sleeve = GND (as Bread Modular V1). Firmware: no PORT_INVEN. Opto-isolated input (ground-loop break, voltage-sensed) is deferred to V2.

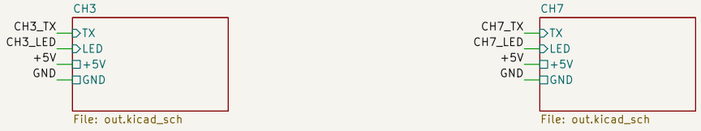
OUTPUTS 1-8: sheets CH1-CH8 (out.kicad_sch): GPIO -> 1k -> TS tip (sleeve GND); LED: GPIO -> 10k -> NPN, +5V -> LED -> 1k -> collector (Bread Modular gate_out)

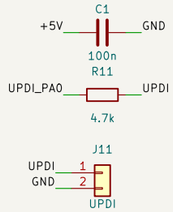


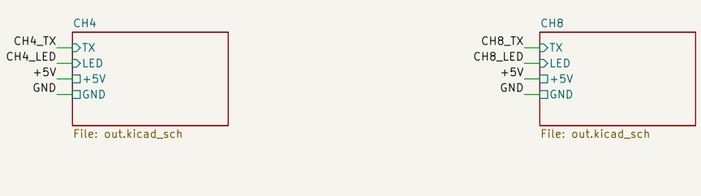
MCU: ATtiny1616 at 5 V, 20 MHz; pin map identical to Bread Modular











UPDI J11: pin 1 UPDI, pin 2 GND (2.54 mm). Power the module from Eurorack while programming.

Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S5 — 5 V logic, LED transistor drivers — not released, not for fabrication

Big Modular

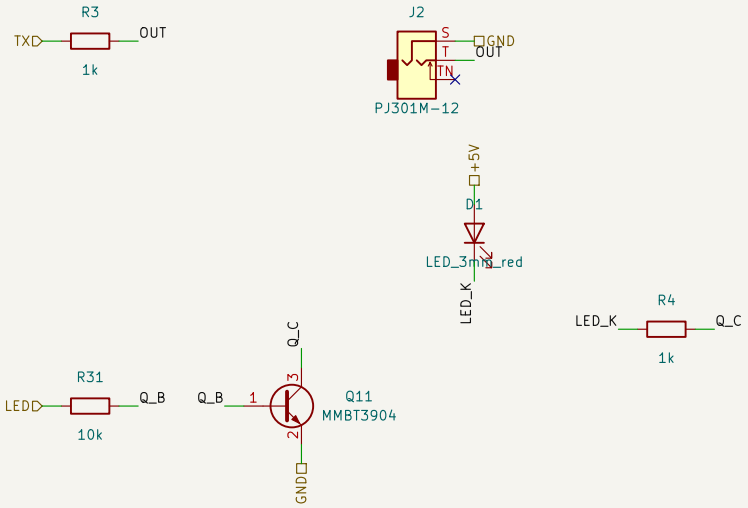
Sheet: /
File: midi.kicad_sch

Title: Big Modular — MIDI

Size: A3	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 1/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

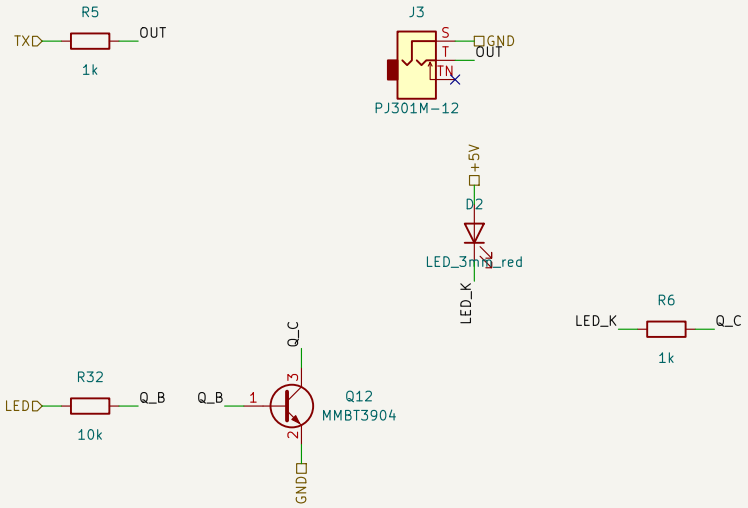
Sheet: /CH1/
File: out.kicad_sch

Title: Big Modular — MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 2/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

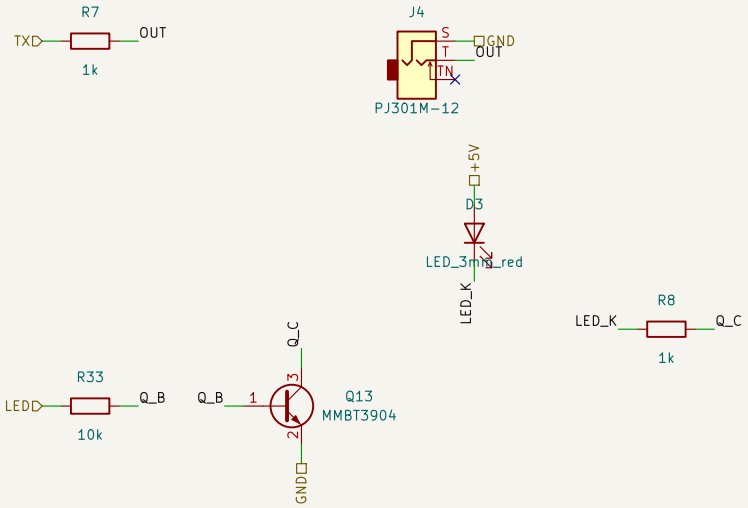
Sheet: /CH2/
File: out.kicad_sch

Title: Big Modular — MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 3/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

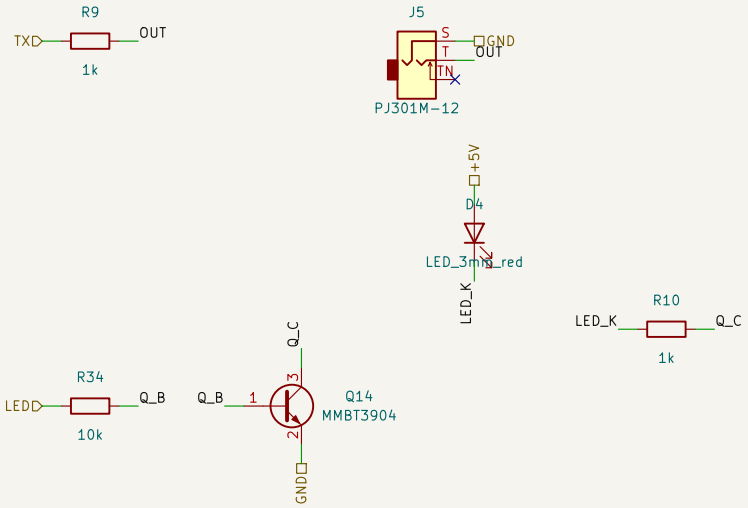
Sheet: /CH3/
File: out.kicad_sch

Title: Big Modular – MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 4/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

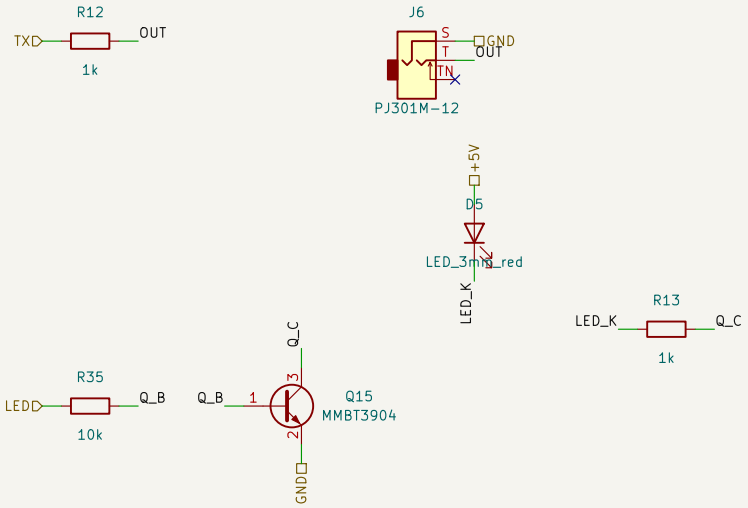
Sheet: /CH4/
File: out.kicad_sch

Title: Big Modular – MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 5/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

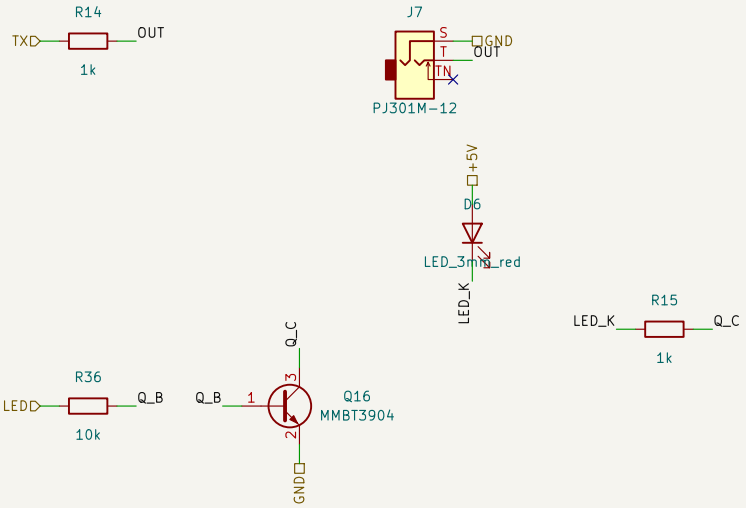
Sheet: /CH5/
File: out.kicad_sch

Title: Big Modular – MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 6/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

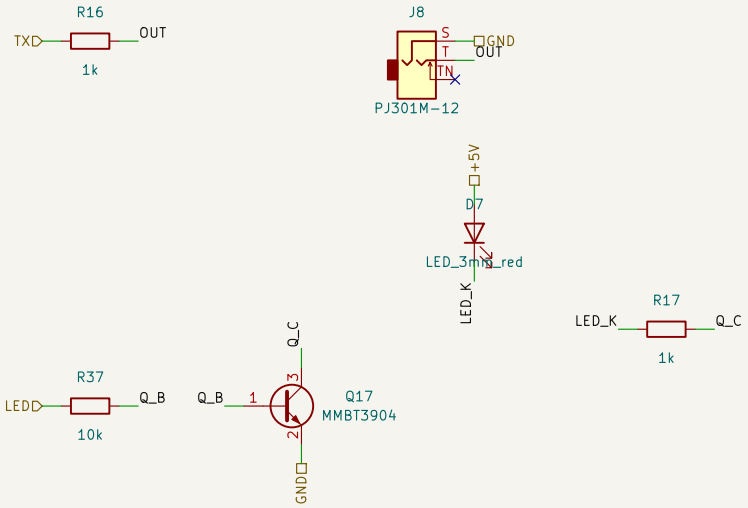
Sheet: /CH6/
File: out.kicad_sch

Title: Big Modular — MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 7/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

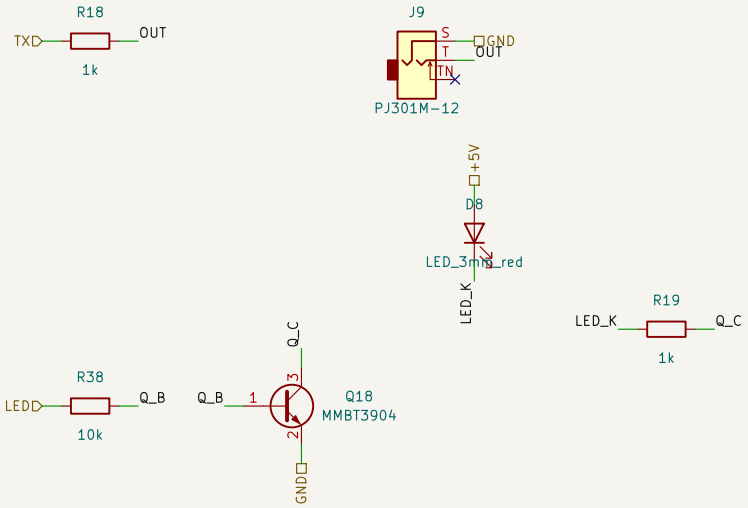
Sheet: /CH7/
File: out.kicad_sch

Title: Big Modular – MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 8/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k –> TS tip, sleeve GND (Bread Modular).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

Sheet: /CH8/
File: out.kicad_sch

Title: Big Modular – MIDI output channel

Size: A4	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 9/9